EXAMINATIONS MODULE (CORE ENGINE)

SmartCampus School Management System
Technical Specification & Implementation Guide — v1.0 • March 2026
Confidential • Internal Documentation

Audience: School Administrators • Teachers • IT Team • Students • Parents
Related Modules: Student Management • Academics • Curriculum • Analytics • Parent Portal

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1. MODULE OVERVIEW

1.1 Purpose

The Examinations Module manages all aspects of exam administration, results processing, and performance analytics. It allows schools to:

- Plan and schedule exam sessions and papers
- Allocate exam seats and resources
- Enter, calculate, and publish student results
- Monitor student performance trends
- Generate transcripts, report cards, and analytics dashboards
- Ensure exam security, integrity, and compliance with curriculum standards

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2. CORE OBJECTIVES

1. Enable creation and management of exam sessions, papers, and schedules
2. Automate result entry, computation, and grade assignment
3. Provide advanced analytics and reporting per student, subject, and class
4. Integrate with Student Management and Academics modules
5. Support exam integrity, secure access, and audit trails
6. Offer role-based dashboards for admins, teachers, and students
7. Ensure scalability for large student populations

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3. REAL-WORLD SYSTEM FLOW

3.1 Lifecycle Flow

Exam Session Created → Papers Assigned → Students Registered → Exam Conducted → Results Entered → Grades Computed → Reports Generated → Parent/Student Notified

3.2 Daily Operational Flow

Administrator/teacher logs in
↓
Create or update exam sessions and papers
↓
Assign students to sessions and allocate seats
↓
Conduct exams (paper or online)
↓
Enter or import results (manual or bulk upload)
↓
System computes grades and assigns competency levels
↓
Results and analytics dashboards updated
↓
Reports sent to parents/students via portal

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4. MODULE ARCHITECTURE

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Examinations Module
│
├── Exam Sessions Management
├── Papers & Question Banks
├── Student Registration & Seat Allocation
├── Result Entry & Computation
├── Grading & Competency Assignment (CBC/8-4-4)
├── Analytics & Performance Dashboards
├── Report Card & Transcript Generation
├── Notifications & Parent/Student Integration
├── Exam Security & Integrity Features
├── Data Import/Export Tools
├── Role-Based Access & Audit Trails
├── Scalability & Performance Management
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5. EXAM SESSIONS MANAGEMENT

5.1 Features

- Create term/annual exam sessions
- Assign papers and subjects to sessions
- Define start/end dates and exam durations
- Track session status (scheduled, ongoing, completed)

5.2 Data Model

ExamSession

- session_id
- name
- term
- start_date
- end_date
- status
- linked_class_ids

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6. PAPERS & QUESTION BANKS

6.1 Features

- Create and upload exam papers
- Link questions to subjects and topics
- Assign marks and grading schemes
- Support randomization for integrity

6.2 Data Model

ExamPaper

- paper_id
- subject_id
- session_id
- questions (array)
- max_score
- grading_scale

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7. STUDENT REGISTRATION & SEAT ALLOCATION

7.1 Features

- Register students for exams automatically via Student Module
- Allocate seats and rooms
- Generate printable seating charts

7.2 Data Model

ExamRegistration

- registration_id
- student_id
- session_id
- paper_id
- seat_number
- room_id

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8. RESULT ENTRY & COMPUTATION

8.1 Features

- Manual entry or bulk import (CSV/Excel/API)
- Auto-calculate totals, averages, and grades
- Assign CBC competency levels (EE, ME, AE, BE)
- Track per-subject, per-class, and per-student performance

8.2 Data Model

ExamResult

- result_id
- student_id
- paper_id
- score
- max_score
- grade
- competency_level
- entered_by
- timestamp

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9. REPORTING & ANALYTICS

9.1 Features

- Subject-level, class-level, and student-level reports
- Charts and visual dashboards for performance trends
- Top performers, at-risk students, and averages
- Printable transcripts and report cards

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10. EXAM SECURITY & INTEGRITY

10.1 Features

- Secure login for teachers and administrators
- Randomization of exam questions
- Plagiarism detection (if online)
- Audit logs for all data entry and edits
- Role-based access control

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11. NOTIFICATIONS & PORTAL INTEGRATION

11.1 Features

- Parent/student portal access for results
- Push/email notifications for released results
- Alerts for missing or incomplete exam entries

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12. MOBILE-FIRST DESIGN

12.1 Requirements

- Mobile-friendly dashboards for teachers and students
- Quick access to exam schedules and results
- Offline caching for limited connectivity

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13. SCALABILITY & PERFORMANCE

- Support thousands of students across multiple campuses
- Handle multiple exam sessions and papers simultaneously
- Optimize queries for rapid result computation and reporting

14. SECURITY & ROLE-BASED ACCESS

Role	Access Scope
Admin	Full access, session creation, reports
Teacher	Assigned papers, enter results, view analytics
Parent	Own child’s results only
Student	Own exam schedule and results only